



Sharif University of Technology
Graduate School of Management and Economics

Master's Thesis in Financial Economics

**Silent Markets in a Noisy World: Tehran's
Stock Exchange Tepid Response to the
Russia-Ukraine War**

By:

Omid Karami

Supervisor:

Dr. Gholamreza Keshavarz Haddad

September 2024

Abstract

This study examines how investors on the Tehran Stock Exchange reacted to the outbreak of the Russia-Ukraine war on February 26, 2022. Utilizing an event-study methodology and daily return data from both the TSE's first and second markets and the Iran Fara Bourse, market behavior was evaluated over a seven-day event window consisting of three trading days before and three trading days after the event date.

The results indicate that, on average, the aggregate Iranian equity market did not exhibit statistically significant abnormal returns compared with many other countries, a result likely related to Iran's relative financial isolation. No significant difference was observed among large-, medium-, and small-cap stock groups, although small firms were affected slightly more. Firms with low-liquidity stocks reacted less than other groups. Industry reactions were heterogeneous, and only the automobile, cement, and food-and-beverage sectors experienced significant price changes. A robustness test based on changing the estimation window also supports the stability of these findings.

Keywords: Russia-Ukraine war; Tehran Stock Exchange; event study; event window; abnormal return; market behavior.

Contents

1	Introduction	1
2	Research Necessity and Research Question	3
3	Literature Review	8
3.1	Events and Financial Markets	8
3.2	Heterogeneous Industry Reactions	9
3.3	Economic Effects of the Russia-Ukraine War	10
4	Data and Summary Statistics	11
5	Research Methodology	15
6	Results	16
6.1	Whole Market	16
6.2	Market Capitalization Criterion	17
6.3	Liquidity Criterion	20
6.4	Industry Criterion	23
7	Robustness Test	29
8	Summary and Conclusion	39
	References	40

List of Tables

1	General structure of the data	12
2	Summary statistics for the main data variables	13
3	Number of firms in each group	13
4	Average abnormal return of the whole equity market during the event window	16
5	Cumulative average abnormal return of the whole equity market during the event window	16
6	Average abnormal return by firm market capitalization during the event window	18
7	Cumulative average abnormal return by firm market capitalization during the event window	19
8	Average abnormal return by stock liquidity during the event window	21
9	Cumulative average abnormal return by stock liquidity during the event window	22
10	Average abnormal return of different industries during the event window	24
11	Cumulative average abnormal return of different industries during the event window	26
12	Average abnormal return of the whole equity market during the event window using a 100-day estimation window	29
13	Cumulative average abnormal return of the whole equity market during the event window using a 100-day estimation window	29
14	Average abnormal return by market capitalization during the event window using a 100-day estimation window	31
15	Cumulative average abnormal return by market capitalization during the event window using a 100-day estimation window	32
16	Average abnormal return by stock liquidity during the event window using a 100-day estimation window	33
17	Cumulative average abnormal return by stock liquidity during the event window using a 100-day estimation window	34

18	Average abnormal return of different industries during the event window using a 100-day estimation window	35
19	Cumulative average abnormal return of different industries during the event window using a 100-day estimation window	37

List of Figures

1	Global crude-oil price trend. Source: investing.com	4
2	Global natural-gas price trend. Source: investing.com	4
3	Global wheat price trend. Source: investing.com	4
4	Global nickel price trend. Source: investing.com	5
5	GPR index over time. Source: matteoiacoviello.com	5
6	Tehran Stock Exchange overall index. Source: tsetmc.com	5
7	Average abnormal return of the whole equity market	17
8	Average abnormal return during the event window by market capitalization	20
9	Average abnormal return during the event window by stock liquidity . . .	23
10	Cumulative average abnormal return of industries over the full event window	28

Chapter 1

Introduction

In the early morning of 24 February 2022, the Russian government launched a military operation against Ukraine under the title of a “special military operation,” which became known in the media as the Russia-Ukraine war. Despite efforts to establish peace between the two countries, the war has continued and has caused extensive damage, including hundreds of thousands of casualties, the displacement of several million civilians to other countries, the destruction of cities and infrastructure, and many other economic and social losses. No short-term prospect for the end of the war is visible.

Beyond its consequences for the two belligerent countries, the war has also affected other countries and the global economy. Russia and Ukraine are among the largest producers of commodities such as agricultural products and energy. The onset of the war disrupted the production and export of many of these commodities, especially in Ukraine. At the same time, Western countries imposed several sanctions on Russia, including restrictions related to trade with that country. These sanctions made the entry of Russian commodities into the global market more difficult. The war also brought Russia and NATO countries, both nuclear powers, closer to the possibility of a full-scale confrontation and therefore increased the level of geopolitical risk in the world.

Iran is both an exporter and an importer of commodities. For example, Iran is one of the major exporters of oil and one of the importers of grain. Therefore, price developments in commodity markets caused by this event may have affected the Iranian economy. In addition, before the war began, Iran had trade relations with both Russia and Ukraine; the war and the sanctions that followed could therefore have influenced Iran’s trade with both countries.

One way to examine the effect of the war on Iran’s economy is to study developments in Iran’s financial market. Financial asset prices reflect investors’ hopes and fears about the future macroeconomy, their willingness to hedge against shocks, and their attempts to earn returns under new conditions. The efficient-market hypothesis states that stock prices rapidly incorporate newly released information. News of events such as the Russia-

Ukraine war is new information for investors and changes their expectations about firms' future profitability according to firm characteristics such as industry, market capitalization, and stock liquidity. Accordingly, by studying price changes in stocks listed on the Tehran Stock Exchange in response to the Russia-Ukraine war, this thesis helps clarify the short-run effects of the war on the Iranian economy from investors' point of view.

The remainder of the thesis is organized as follows. Chapter 2 states the research question and explains the importance of examining it. Chapter 3 reviews the related literature. Chapter 4 describes the data and summary statistics. Chapter 5 sets out the research methodology. Chapter 6 presents the empirical results. Chapter 7 reports robustness tests, and Chapter 8 concludes.

Chapter 2

Research Necessity and Research Question

Before stating the research question, it is useful to consider the position of Russia and Ukraine in commodity markets before the war. Russia was the world's largest wheat exporter, accounting for 18 percent of global wheat exports. It was also the largest exporter of natural gas (25 percent), palladium (23 percent), nickel (22 percent), and chemical fertilizers (14 percent). Russia accounted for 18 percent of global coal exports, 14 percent of platinum exports, 11 percent of crude-oil exports, and 10 percent of refined-aluminum exports. Ukraine, in addition to its 7 percent share in world wheat exports, was the largest exporter of seed oil, a product mainly used in cooking, and supplied two-fifths of global output. It was also the fourth-largest exporter of corn, with 13 percent of world exports. Ukraine also produced 50 percent of the world's neon gas, an important input in chip production.

With the outbreak of the war, economic activity in many parts of Ukraine became practically impossible. Ukrainian ports on the Black Sea, which had been used especially for exporting domestic production, lost much of their functionality. Russia did not face the same operational disruptions because of its stronger military position, but Western countries led by the United States imposed severe sanctions against it. These sanctions covered trade in many goods, including commodities, the disconnection of Russian banks from the international banking system, and other restrictions. The beginning of the war and the announcement of sanctions against Russia triggered price reactions in many commodity markets.

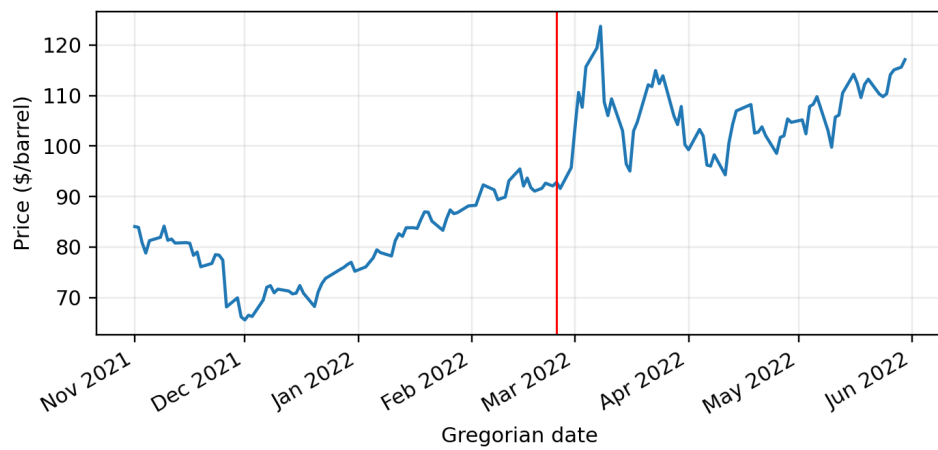


Figure 1: Global crude-oil price trend. Source: investing.com

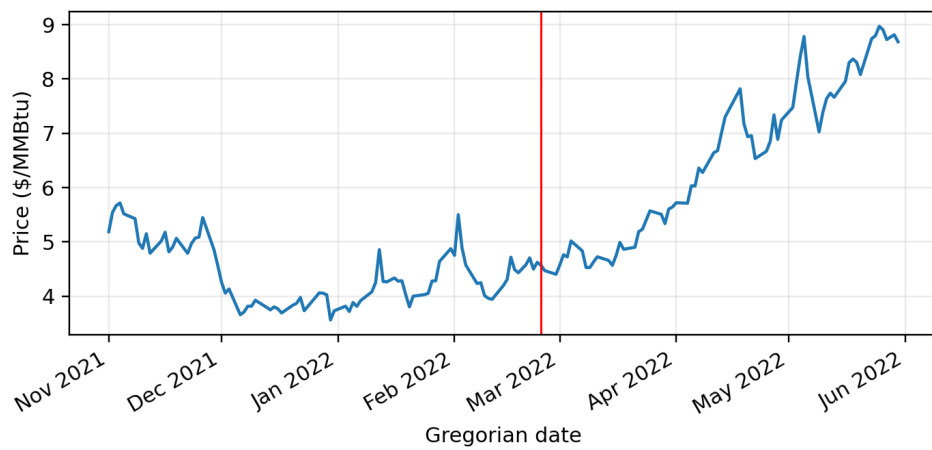


Figure 2: Global natural-gas price trend. Source: investing.com

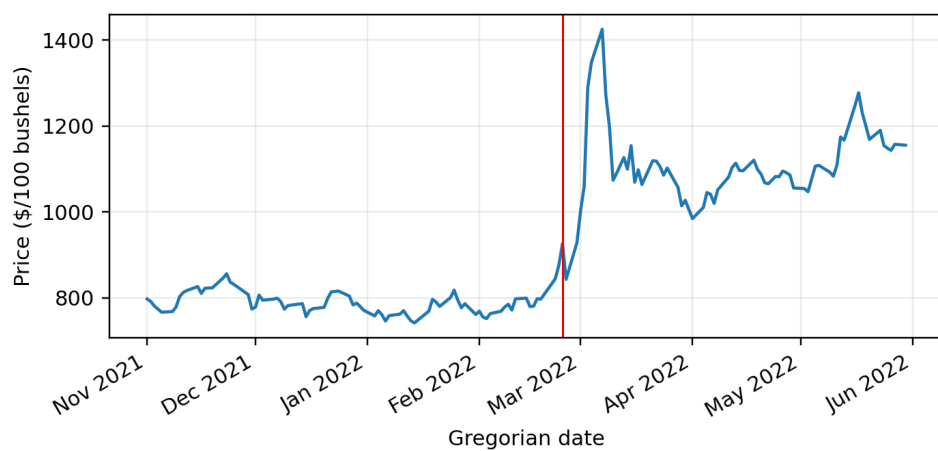


Figure 3: Global wheat price trend. Source: investing.com

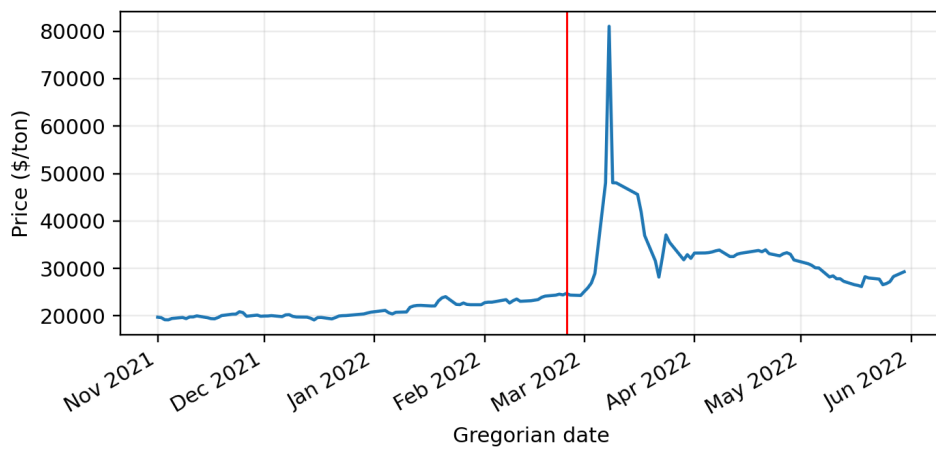


Figure 4: Global nickel price trend. Source: investing.com

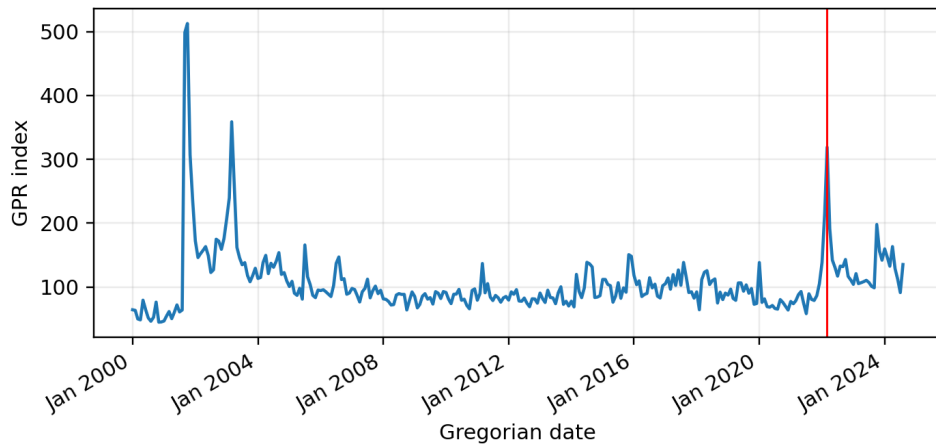


Figure 5: GPR index over time. Source: matteoiacoviello.com

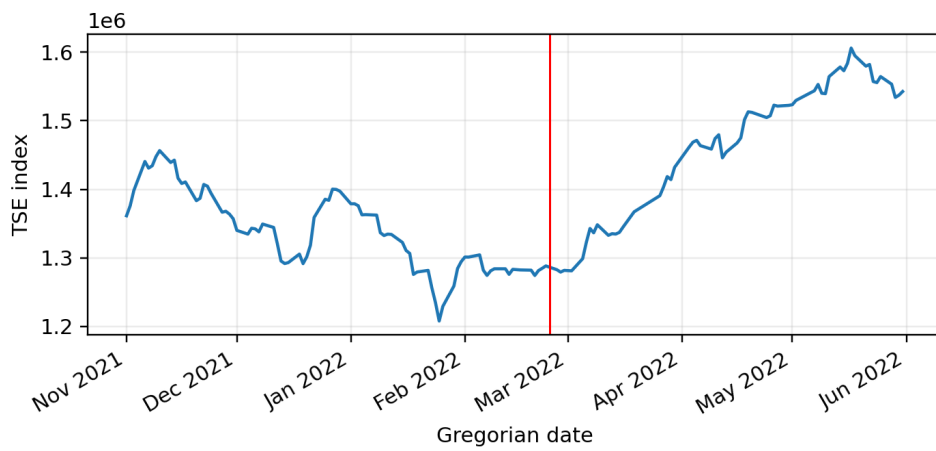


Figure 6: Tehran Stock Exchange overall index. Source: tsetmc.com

Figure 1 shows that the global price of crude oil rose sharply around the beginning of the war and even reached 123 dollars per barrel on one day. Figure 2 shows that natural gas, like oil, rose after the war began; however, unlike oil, it did not react immediately. One reason is that at the beginning of the war, European countries did not accompany the United States in sanctioning Russian gas in the same way that they sanctioned Russian oil. Moreover, markets initially did not take Russian threats to cut gas supplies to Europe seriously. Later, as Russian threats became more serious and Europe attempted to reduce gas consumption and find substitute sources, the natural-gas market also reacted. Wheat and nickel, as agricultural and basic-metal commodities, clearly experienced price jumps at the beginning of the war. Part of these price increases can be attributed to rising demand for commodities as COVID-19 restrictions eased and the global economy emerged from the pandemic recession; nevertheless, the full increase, especially the jumps at the beginning of the war, cannot be explained by that factor alone.

The war also increased the level of geopolitical risk. This increase is visible in the GPR index. The GPR index introduced by Caldara and Iacoviello measures the level of geopolitical risk in the world. According to this index, in February and March 2022 the level of geopolitical risk rose sharply because of the Russia-Ukraine war. Such increases in risk naturally affect economic activity and, in turn, financial markets.

Iran, as an important trade partner of Russia, may have been affected by the war. Trade between Iran and Russia in 2021 was close to 4.035 billion dollars, which indicates the importance of this relationship. On one hand, Iran may have benefited from rising prices for commodities such as oil and copper. On the other hand, sanctions against Russia led that country to sell some of its products in informal markets at larger discounts, which could affect Iran's market share. In addition, Iran is a large importer of wheat and corn and may have been harmed by increases in the prices of these products. The global inflationary pressure created by the war could also have negative effects on Iran's economy.

As noted in the introduction, reactions in financial markets can be used to study the effects of events on an economy. This is also true for Iran. By examining the reaction of the Tehran equity market, we can infer how investors and market participants evaluated the effects of the Ukraine war on Iranian firms. Still, the effect of the Russia-Ukraine war on the Tehran Stock Exchange is not obvious. Rising commodity prices and geopolitical risk may affect commodity-oriented industries such as petrochemicals, basic metals, and agriculture. By contrast, the particular structure of Iran's capital market, international sanctions against Iran, and domestic constraints may have made the event less influential for Tehran-listed stocks. In other words, because of its special economic and political conditions, the Iranian stock market may react less to global events.

Studying only the aggregate market may not provide a complete picture of the war's effects. Therefore, this thesis also examines three dimensions: (1) firms' industry of activ-

ity, because commodity-based industries may be affected differently; (2) stock liquidity, because more liquid stocks may react faster to global news while less liquid stocks may adjust with delay; and (3) firm position by market capitalization, because larger firms may have better access to financial resources and may adapt more easily to crisis conditions, whereas smaller firms may be more vulnerable.

The central aim of this research is to evaluate the effect of the Russia-Ukraine war on stock returns of firms listed on the Tehran Stock Exchange and to determine whether the event generated abnormal returns relative to normal market conditions. The thesis also examines whether stock liquidity, industry, and market capitalization changed the price response to the event. Answering these questions is scientifically important and may also be useful for investors and policymakers. The results can help investors diversify portfolios to hedge against war-related risk and can help policymakers better understand how global shocks are transmitted to Iran's economy.

Chapter 3

Literature Review

Firms need financial resources to invest and expand their economic activities. One source of financing is the capital market, where companies can offer shares. Stocks are among the main financial assets traded in financial markets, and their prices are determined by the interaction of supply and demand. Anything that changes these forces can change stock prices.

The empirical literature on efficient markets shows that prices respond rapidly to new information. Fama, Fisher, Jensen, and Roll (1969) document that stock prices adjust quickly to information related to stock splits, and Fama (1970) reviews evidence that historical price information has little power for forecasting future returns. With the expansion of communication technologies and instant access to news, financial-market participants can rapidly obtain information about events around the world. Niederhoffer (1971) shows that major world events reflected in newspaper headlines can provide useful information about potential stock-price changes and that markets may overreact to bad news. Fair (2002) studies large changes in U.S. equity prices using S&P 500 futures and links many of those changes to identifiable news events. Kaplanski and Levy (2010) show that aviation disasters influence investor sentiment and can lead to mispricing.

This chapter first reviews studies that examine the effect of different events on financial markets. It then reviews work on heterogeneous industry reactions to events. Finally, because the event studied here is the Russia-Ukraine war, it reviews research on the economic effects of that war.

3.1 Events and Financial Markets

The COVID-19 outbreak was one of the most important recent events affecting the economies of all countries. Pandey and Kumari (2021), using an event-study approach and data from 49 equity indices, show that COVID-19 had a negative and significant effect on global

stock markets and that developed markets were affected for a longer period than emerging markets. Lyocsa et al. (2020) study uncertainty in financial markets created by COVID-19 and show that Google search intensity for coronavirus-related terms had predictive content for realized volatility and price changes in stock markets.

Terrorist events are another category of events that can affect financial markets. Chesney, Reshetar, and Karaman (2011) examine the effect of terrorism on stock, bond, and commodity markets using event studies, non-parametric methods, and filtered GARCH-EVT methods. They find that about two-thirds of the terrorist attacks considered had a negative and significant effect on at least one country's stock market. Airline and insurance stocks were especially sensitive, whereas banking stocks were less sensitive. Chen and Siems (2007) examine global financial-market reactions to terrorist attacks and military conflicts and find that the U.S. capital market became more resilient over time and recovered from terrorist shocks more quickly than many other markets.

Several studies focus on wars and political conflicts. Abadie and Gardeazabal (2003) estimate the economic costs of terrorism in the Basque Country using a synthetic-control method and show that conflict reduced GDP relative to a synthetic no-terrorism counterfactual. Guidolin and La Ferrara (2010) show that violent conflict can have economically important effects on asset prices. Rigobon and Sack (2005) study war risk and U.S. financial markets and find that increases in war risk reduced stock prices and increased oil prices. Choudhry (2010), Hudson and Urquhart (2015), and other studies show that major war-related events can influence stock-market returns.

Political uncertainty and geopolitical risk also affect financial markets. Mei and Guo (2004), Berkman, Jacobsen, and Lee (2011), Chau, Deesomsak, and Wang (2014), Dimic, Orlov, and Piljak (2015), and Boutchkova et al. (2012) document links between political risk, disaster risk, volatility, and stock returns. Studies such as Bash and Alsaifi (2019) and Buigut and Kapar (2020) use event-study methods to examine regional political and diplomatic shocks.

3.2 Heterogeneous Industry Reactions

A major theme in the literature is that industries do not react uniformly to events. Terrorism, wars, health crises, and diplomatic disputes affect sectors through different channels. For example, airlines, insurance, banking, energy, metals, food, and transportation may be exposed differently because they differ in input structure, demand sensitivity, regulation, exposure to international prices, and investor attention. Wang and Liu (2022) show that the psychological and industrial effects of COVID-19 differed across Chinese firms. Hassan et al. (2022) document heterogeneous sectoral returns around border disputes. This line of research motivates the industry-level analysis in this thesis.